

KOM4521 Multivariable Control Theory

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Syllabus:

Week 7-9-10

- Calculation of Controllability and Observability Grammians
- PBH Test for Controllability and Observability

Controllability Grammian

Continuous-time controllability Gramian:

If a continuous-time system is controllable, then

$$W_c(t) = \int_0^t e^{A\tau} B B^T e^{A^T \tau} d\tau$$

is nonsingular for $t > 0$.

Controllability Grammian

Consider

$$x(t_1) = e^{At_1}x(0) + \int_0^{t_1} e^{A(t_1-\tau)} Bu(\tau) d\tau.$$

We claim that for any $x(0) = x_0$ and any $x(t_1) = x_1$ the input

$$u(t) = -B^T e^{A^T(t_1-t)} W_c^{-1}(t_1) [e^{At_1}x_0 - x_1]$$

will transfer x_0 to x_1 at time t_1 .

Controllability Grammian

PROOF: Substitute the expression for $u(t)$ into the convolution expression:

$$\begin{aligned}x(t_1) &= e^{At_1}x(0) - \int_0^{t_1} e^{A(t_1-\tau)} BB^T e^{A^T(t_1-\tau)} d\tau W_c^{-1}(t_1) [e^{At_1}x_0 - x_1] \\&= e^{At_1}x(0) - \int_0^{t_1} e^{A\beta} BB^T e^{A^T\beta} d\beta W_c^{-1}(t_1) [e^{At_1}x_0 - x_1] \\&= e^{At_1}x(0) - W_c(t_1)W_c^{-1}(t_1) [e^{At_1}x_0 - x_1] \\&= e^{At_1}x(0) - e^{At_1}x_0 + x_1 = x_1.\end{aligned}$$

Controllability Grammian

Therefore, we can compute the input $u(t)$ required to transfer the state of the system from one state to another over an arbitrary interval of time. The solution is also the minimum-energy solution.

Controllability Grammian

Example: Consider the system in diagonal form

$$\dot{x}(t) = \begin{bmatrix} -0.5 & 0 \\ 0 & -1 \end{bmatrix} x(t) + \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} u(t).$$

Controllability matrix is:

$$\mathcal{C} = \begin{bmatrix} 0.5 & -0.25 \\ 1 & -1 \end{bmatrix}$$

which has rank 2, so the system is controllable.

Controllability Grammian

Consider the input required to move the system state from $x(0) = [10 \ -1]^T$ to zero in two seconds.

$$\begin{aligned} W_c(2) &= \int_0^2 \left(\begin{bmatrix} e^{-0.5\tau} & 0 \\ 0 & e^{-\tau} \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} \begin{bmatrix} 0.5 & 1 \end{bmatrix} \begin{bmatrix} e^{-0.5\tau} & 0 \\ 0 & e^{-\tau} \end{bmatrix} \right) d\tau \\ &= \begin{bmatrix} 0.2162 & 0.3167 \\ 0.3167 & 0.4908 \end{bmatrix}, \end{aligned}$$

and

$$\begin{aligned} u(t) &= - \begin{bmatrix} 0.5 & 1 \end{bmatrix} \begin{bmatrix} e^{-0.5(2-t)} & 0 \\ 0 & e^{-(2-t)} \end{bmatrix} W_c(2)^{-1} \begin{bmatrix} e^{-1} & 0 \\ 0 & e^{-2} \end{bmatrix} \begin{bmatrix} 10 \\ -1 \end{bmatrix} \\ &= -58.82e^{0.5t} + 27.96e^t. \end{aligned}$$

Controllability Grammian

If a continuous-time system is controllable, and if it is also stable, then

$$W_c = \int_0^{\infty} e^{A\tau} B B^T e^{A^T \tau} d\tau$$

can be found by solving for the unique (positive-definite) solution to the (Lyapunov) equation

$$A W_c + W_c A^T = -B B^T.$$

W_c is called the controllability Gramian.

Observability Grammian

Continuous-time Observability Gramian:

If a continuous-time system is observable and $W_o(t)$ is nonsingular for $t > 0$.

$$W_o(t) = \int_0^t e^{A^T \tau} C^T C e^{A \tau} d\tau.$$

Observability Grammian

We can prove that

$$x(0) = W_o^{-1}(t_1) \int_0^{t_1} e^{A^T t} C^T \bar{y}(t) dt$$

where

$$\bar{y}(t) = y(t) - C \int_0^t e^{A(t-\tau)} B u(\tau) d\tau - D u(t) = C e^{At} x(0).$$

Therefore, we can determine the initial state $x(0)$ given a finite observation period (and not use differentiators!).

Observability Grammian

PROOF: We prove that the above equations are correct by substitution:

$$\begin{aligned} W_o^{-1}(t_1) \int_0^{t_1} e^{A^T t} C^T \bar{y}(t) dt &= W_o^{-1}(t_1) \int_0^{t_1} e^{A^T t} C^T C e^{At} dt x(0) \\ &= W_o^{-1}(t_1) W_o(t_1) x(0) \\ &= x(0). \end{aligned}$$

Observability Grammian

- If a continuous-time system is observable, and if it is also stable, then

$$W_o = \int_0^{\infty} e^{A^T \tau} C^T C e^{A \tau} d\tau$$

is the unique (positive-definite) solution to the (Lyapunov) equation

$$A^T W_o + W_o A = -C^T C.$$

W_o is called the observability Gramian.

Transformation to controllability form

Given a system

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t),$$

can we find a transformation (similarity) matrix T to transform this system into controllability form? Recall, this looks like:

$$\dot{x}_{co}(t) = \begin{bmatrix} 0 & \cdots & 0 & -a_n \\ 1 & & \vdots & -a_{n-1} \\ \vdots & \ddots & & \vdots \\ 0 & \cdots & 1 & -a_1 \end{bmatrix} x_{co}(t) + \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} \beta_1 & \beta_2 & \cdots & \beta_n \end{bmatrix} x_{co}(t) + Du(t).$$

Transformation to controllability form

- Note that $C_{co} = I$ so it is controllable. Thus, our original system must be controllable.
- The transformation is accomplished via

$$x = T x_{co}.$$

- Thus

$$T^{-1}AT = A_{co}, \quad T^{-1}B = B_{co}$$

$$CT = C_{co}, \quad D = D_{co}.$$

- Let's find T explicitly; let

$$T = [t_1, \dots, t_n].$$

Transformation to controllability form

- Note that $B = TB_{co}$ or

$$B = T \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = t_1.$$

- So, $t_1 = B$. From $AT = TA_{co}$ we have

$$A[t_1, \dots, t_n] = [t_1, \dots, t_n] \begin{bmatrix} 0 & \cdots & 0 & -a_n \\ 1 & & \vdots & -a_{n-1} \\ \vdots & \ddots & & \vdots \\ 0 & \cdots & 1 & -a_1 \end{bmatrix}.$$

- So

$$[At_1, \dots, At_n] = \left[t_2, \dots, t_n, -\sum_{k=1}^n a_k t_{n-k+1} \right].$$

Transformation to controllability form

- By induction,

$$At_1 = t_2 = AB$$

$$At_2 = t_3 = A^2B, \text{ and so forth } \dots$$

so

$$T = [B \ AB \ \dots \ A^{n-1}B] = C.$$

Transformation to controllability form

A system $\{A, B, C, D\}$ can be transformed to controllability canonical form if and only if it is controllable, in which case the change of coordinates is

$$x = \mathcal{C}x_{co}.$$

Transformation to controllability form

EXTENSION I: If $x_{\text{old}} = T x_{\text{new}}$ then $T = C_{\text{old}} C_{\text{new}}^{-1}$. That is, to convert between any two realizations, T is a combination of the controllability matrices of the two different realizations.

$$\begin{aligned} C_{\text{new}} &= [B_{\text{new}} \quad A_{\text{new}} B_{\text{new}} \quad \cdots \quad A_{\text{new}}^{n-1} B_{\text{new}}] \\ &= [T^{-1} B_{\text{old}} \quad T^{-1} A_{\text{old}} T T^{-1} B_{\text{old}} \quad \cdots \quad T^{-1} A_{\text{old}}^{n-1} T T^{-1} B_{\text{old}}] \\ &= T^{-1} C_{\text{old}}, \end{aligned}$$

or

$$T = C_{\text{old}} C_{\text{new}}^{-1}.$$

Transformation to observability form

EXTENSION II: If $x_{\text{old}} = T x_{\text{new}}$ then $T = \mathcal{O}_{\text{old}}^{-1} \mathcal{O}_{\text{new}}$. This can be shown in a similar way.

Canonical (Kalman) decompositions

What happens if $\{A, B\}$ not controllable or if $\{A, C\}$ not observable?

- Is “part” of the system controllable?
- Is “part” of the system observable?

Canonical (Kalman) decompositions

Given a system with $\{A, B\}$ not controllable,

$$\dot{x}(t) = Ax(t) + Bu(t),$$

$$y(t) = Cx(t)$$

Let's try to transform to controllability canonical form.

Let $t_1 = B$; $t_2 = AB$, ... , $t_r = A^{r-1}B$, and suppose $t_1, t_2, \dots, t_r$ are independent.

Canonical (Kalman) decompositions

But $t_{r+1} = A^r B = -\alpha_r t_1 - \cdots - \alpha_1 t_r$

for some constants $\alpha_1, \alpha_2, \dots, \alpha_r$.

Then $\text{rank}(C) = r$ since the vectors $A^r B, A^{r+1} B, \dots, A^{n-1} B$ can all be expressed as a linear combination of $t_1, t_2, \dots, t_r$.

Let $s_r, \dots, s_n$ be your favorite vectors for which $\bar{C}_{\text{old}} = \begin{bmatrix} t_1 & \cdots & t_r & \vdots & s_{r+1} & \cdots & s_n \end{bmatrix}$ is invertible.

Canonical (Kalman) decompositions

- If we were able to change coordinates to controllability form via $x_{\text{old}} = T x_{\text{new}}$, we would use $T = C_{\text{old}} C_{\text{new}}^{-1}$.
- But, the system is not controllable, so we use $T = \bar{C}_{\text{old}} C_{\text{new}}^{-1} = \bar{C}_{\text{old}}$.

$$\begin{bmatrix} \dot{x}_c(t) \\ \dot{x}_{\bar{c}}(t) \end{bmatrix} = \begin{bmatrix} A_c & A_{12} \\ 0 & A_{\bar{c}} \end{bmatrix} \begin{bmatrix} x_c(t) \\ x_{\bar{c}}(t) \end{bmatrix} + \begin{bmatrix} B_c \\ 0 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} C_c & C_{\bar{c}} \end{bmatrix} \begin{bmatrix} x_c(t) \\ x_{\bar{c}}(t) \end{bmatrix} + Du(t).$$

- A_c is a right-companion matrix and B_c is of the controllability-canonical form.
- We see that the uncontrollable modes $x_{\bar{c}}$ are completely decoupled from $u(t)$.

Canonical (Kalman) decompositions

Example: Consider $\frac{1}{s+1} = \frac{s-1}{(s+1)(s-1)} = \frac{s-1}{s^2-1}$.

In observer-canonical form,

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ -1 \end{bmatrix} u(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t).$$

$$t_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{and} \quad t_2 = At_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -t_1.$$

Canonical (Kalman) decompositions

So $\text{rank}(\mathcal{C}) = 1$. Let $s_1 = [1 \ 0]^T$. The converted state-space form is

$$\begin{aligned}\dot{\bar{x}}(t) &= \begin{bmatrix} -1 & -1 \\ 0 & 1 \end{bmatrix} \bar{x}(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 1 & 1 \end{bmatrix} \bar{x}(t).\end{aligned}$$

Now suppose that we desire to transform a system that is not observable. The dual form separates out the unobservable states.

Canonical (Kalman) decompositions

Let $t_1 = C$, $t_2 = CA, \dots$, $t_r = CA^{r-1}$, and suppose $t_1, t_2, \dots, t_r$ are independent but

$$t_{r+1} = CA^r = -\alpha_r t_1 - \dots - \alpha_1 t_r$$

for some constants $\alpha_1, \alpha_2, \dots, \alpha_r$.

Then, let $s_{r+1}, \dots, t_n$ be your favorite row vectors for which is invertible.

$$\bar{O}_{\text{old}} = \begin{bmatrix} t_1 \\ \vdots \\ t_r \\ \hline s_{r+1} \\ \vdots \\ s_n \end{bmatrix}$$

Canonical (Kalman) decompositions

- If we were able to change coordinates to observability form via $x_{\text{old}} = T x_{\text{new}}$, we would use $T = \mathcal{O}_{\text{old}}^{-1} \mathcal{O}_{\text{new}}$.
- But, the system is not observable, so we use $T = \bar{\mathcal{O}}_{\text{old}}^{-1} \mathcal{O}_{\text{new}} = \bar{\mathcal{O}}_{\text{old}}^{-1}$.

Then, we get the Kalman decomposition

$$\begin{bmatrix} \dot{x}_o(t) \\ \dot{x}_{\bar{o}}(t) \end{bmatrix} = \begin{bmatrix} A_o & 0 \\ A_{21} & A_{\bar{o}} \end{bmatrix} \begin{bmatrix} x_o(t) \\ x_{\bar{o}}(t) \end{bmatrix} + \begin{bmatrix} B_o \\ B_{\bar{o}} \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} C_o & 0 \end{bmatrix} \begin{bmatrix} x_o(t) \\ x_{\bar{o}}(t) \end{bmatrix} + Du(t).$$

Popov–Belevitch–Hautus (PBH) Test

PBH Eigenvector Test:

$\{C, A\}$ is an unobservable pair iff a non-zero eigenvector v of A satisfies $Cv = 0$.

→ Suppose $Av = \lambda v$ and $Cv = 0$ for $v \neq 0$. Then $CAv = \lambda Cv = 0$ and so forth up to $CA^{n-1}v = \lambda^{n-1}Cv = 0$. So, $\mathcal{O}v = 0$ and since $v \neq 0$ this means that $\mathcal{O}$ is not full rank and $\{C, A\}$ is not observable.

Popov–Belevitch–Hautus (PBH) Test

← Now suppose that $\mathcal{O}$ is not full rank ($\{C, A\}$ unobservable). Let's extract the unobservable part. That is, find T such that

$$T^{-1}AT = \begin{bmatrix} A_o & 0 \\ A_{21} & A_{\bar{o}} \end{bmatrix}$$

$$CT = \begin{bmatrix} C_o & 0 \end{bmatrix},$$

where A_o is size r where $r = \text{rank}(\mathcal{O})$ and therefore $A_{\bar{o}}$ is size $(n - r)$.

Popov–Belevitch–Hautus (PBH) Test

Let $v_2 \neq 0$ be an eigenvector of $A_{\bar{o}}$. Then

$$T^{-1}AT \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = \begin{bmatrix} A_o & 0 \\ A_{21} & A_{\bar{o}} \end{bmatrix} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = \lambda \begin{bmatrix} 0 \\ v_2 \end{bmatrix}$$

so we have found an eigenvector z of A

$$Az = \lambda z \quad \text{where} \quad z = T \begin{bmatrix} 0 \\ v_2 \end{bmatrix}.$$

Now, we just need to show that $Cz = 0$
(note: $z \neq 0$).

Popov–Belevitch–Hautus (PBH) Test

$$Cz = \underbrace{CT}_{\tilde{C}} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = \begin{bmatrix} C_o & 0 \end{bmatrix} \begin{bmatrix} 0 \\ v_2 \end{bmatrix} = 0$$

DUAL: $\{A, B\}$ is uncontrollable iff there is a left eigenvector ω^T of A such that $\omega^T B = 0$.

Popov–Belevitch–Hautus (PBH) Test

In modal coordinates, homogeneous response is

$$x(t) = \sum_{i=1}^n e^{\lambda_i t} v_i (w_i^T x(0)) \quad \text{and} \quad y(t) = C x(t).$$

$$y(t) = \sum_{i=1}^n e^{\lambda_i t} C v_i (w_i^T x(0)).$$

- If $\{C, A\}$ is unobservable, then it has an unobservable mode, where $A v_i = \lambda_i v_i$ and $C v_i = 0$.

Popov–Belevitch–Hautus (PBH) Test

- If $\{A, B\}$ is uncontrollable, then it has an uncontrollable mode, namely the coefficients of the state along that mode is independent of the input $u(t)$.

The coefficients of x in the mode associated with λ are $w^T x$.

$$\frac{d}{dt}(w^T x) = w^T (Ax + Bu) = \lambda(w^T x)$$

$$w^T x(t) = e^{\lambda t} (w^T x(0))$$

regardless of the input $u(t)$!

Popov–Belevitch–Hautus (PBH) Test

PBH Rank Test:

□ $\{A, B\}$ controllable iff $\text{rank} \begin{bmatrix} sI - A & B \end{bmatrix} = n$ for all $s \in \mathbb{C}$.

→ If $\text{rank} \begin{bmatrix} sI - A & B \end{bmatrix} = n$ for all $s \in \mathbb{C}$. then there can be no nonzero vector v such that

$$v^T \begin{bmatrix} sI - A & B \end{bmatrix} = \begin{bmatrix} v^T(sI - A) & v^T B \end{bmatrix} = 0.$$

Consequently, there is no nonzero vector v such that $v^T s = v^T A$ and $v^T B = 0$. By the PBH eigenvector test, the system will therefore be controllable.

Popov–Belevitch–Hautus (PBH) Test

← If the system is controllable, then there is no nonzero vector v such that $v^T s = v^T A$ and $v^T B = 0$ by the PBH eigenvector test.

Therefore, $\text{rank} \begin{bmatrix} sI - A & B \end{bmatrix} = n$ for all $s \in \mathbb{C}$.

Popov–Belevitch–Hautus (PBH) Test

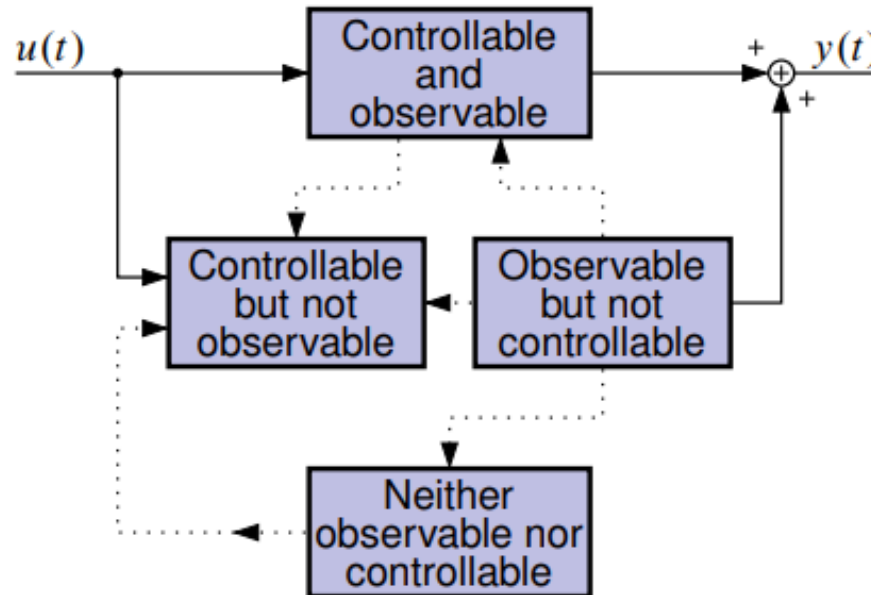
□ $\{C, A\}$ observable iff $\text{rank} \begin{bmatrix} C \\ sI - A \end{bmatrix} = n$ for all $s \in \mathbb{C}$.

Proof similar.

$\text{rank} \begin{bmatrix} sI - A & B \end{bmatrix} = n$ for all s not eigenvalues of A , so the test is really $\{A, B\}$ controllable iff $\text{rank} \begin{bmatrix} \lambda_i I - A & B \end{bmatrix} = n$ for λ_i , $i = 1, \dots, n$ the eigenvalues of A . (Dual argument for observability).

- If $\begin{bmatrix} sI - A & B \end{bmatrix}$ drops rank at $s = \lambda$ then there is an uncontrollable mode with exponent (frequency) λ .
- If $\begin{bmatrix} C \\ sI - A \end{bmatrix}$ drops rank at $s = \lambda$ then there is an unobservable mode with exponent (frequency) λ .

Popov–Belevitch–Hautus (PBH) Test



- ❑ **STABILIZABLE**: A system whose unstable modes are controllable
- ❑ **DETECTABLE**: A system whose unstable modes are observable

Minimal Realization

- To realize $H(s) = \frac{1}{s+1}$ we could use

1. $\dot{x}(t) = -x(t) + u(t)$, $y(t) = x(t)$. This gives

$$A = [-1], B = [1], C = [1], D = [0].$$

$$\frac{C \operatorname{adj}(sI - A) B}{\det(sI - A)} = \frac{1}{s+1}.$$

This realization is both controllable and observable.

Minimal Realization

2. Observer realization of $\frac{1}{s+1} \frac{s-1}{s-1} = \frac{s-1}{s^2-1}$. This gives

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix}, D = [0].$$

$$\frac{C \operatorname{adj}(sI - A) B}{\det(sI - A)} = \frac{s-1}{s^2-1} = \frac{1}{s+1}.$$

$$\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \text{ Observable.}$$

$$\mathcal{C} = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}. \text{ Not controllable.}$$

Minimal Realization

3. Controller realization of $\frac{1}{s+1} \frac{s+10}{s+10} = \frac{s+10}{s^2+11s+10}$.

$$A = \begin{bmatrix} -11 & -10 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 10 \end{bmatrix}, D = [0].$$

$$\frac{C \operatorname{adj}(sI - A) B}{\det(sI - A)} = \frac{s+10}{s^2+11s+10} = \frac{1}{s+1}.$$

$$C = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 1 & -11 \\ 0 & 1 \end{bmatrix}. \text{ Controllable.}$$

$$\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 10 \\ -1 & -10 \end{bmatrix}. \text{ Not observable.}$$

Minimal Realization

Non-minimal realizations of transfer functions will either be uncontrollable, unobservable, or both. Four equivalent statements:

- I. There exist common roots of $C \operatorname{adj}(sI - A) B$ and $\det(sI - A)$.
- II. There exist eigenvalues of A which are not poles of $G(s)$, counting multiplicities.

Minimal Realization

III. The system is either unobservable or uncontrollable.

IV. There exist extra (unnecessary) states—non minimal.

Minimal Realization

PROOF: I $\Longleftrightarrow$ II

I $\Rightarrow$ II: The transfer function

$$\begin{aligned} G(s) &= C(sI - A)^{-1}B + D \\ &= \frac{C \operatorname{adj}(sI - A)B + D \det(sI - A)}{\det(sI - A)}. \end{aligned}$$

If there are common roots in $C \operatorname{adj}(sI - A)B$ and $\det(sI - A)$ they will cancel out of the transfer function. But, eigenvalues of A are $\det(sI - A) = 0$, so poles of $G(s)$ will not contain all eigenvalues of A .

II $\Rightarrow$ I: Eigenvalues of A are $\det(sI - A) = 0$. Poles of $G(s)$, from above, are $\det(sI - A) = 0$ unless canceled. The only way to cancel a pole is to have common root in $C \operatorname{adj}(sI - A)B$.

Minimal Realization

PROOF: I $\iff$ IV $\{A, B, C, D\}$ is minimal iff $C \operatorname{adj}(sI - A)B$ and $\det(sI - A)$ are *coprime* (have no common roots).

I $\implies$ IV: Suppose $C \operatorname{adj}(sI - A)B$ and $\det(sI - A)$ have common roots.

- Cancel to get

$$G(s) = \frac{C \operatorname{adj}(sI - A)B}{\det(sI - A)} + D = \frac{b_r(s)}{a_r(s)}$$

where $b_r(s)$ and $a_r(s)$ are coprime (“*r*” means “reduced”).

- Because of cancellation, $k = \deg(a_r) < \deg(\det(sI - A)) = n$.
- Consider controller canonical form realization of $b_r(s)/a_r(s)$, for example.
- It has k states, but same transfer function as $\{A, B, C, D\}$, contradicting that $\{A, B, C, D\}$ minimal.

Minimal Realization

IV $\implies$ I: If there are extra states (non-minimal) then $n > k$ ($n = \#$ states, $k = \#$ poles in $G(s)$).

$$\begin{aligned} G(s) &= C(sI - A)^{-1}B + D \\ &= \frac{C \operatorname{adj}(sI - A)B + D \det(sI - A)}{\det(sI - A)} \end{aligned}$$

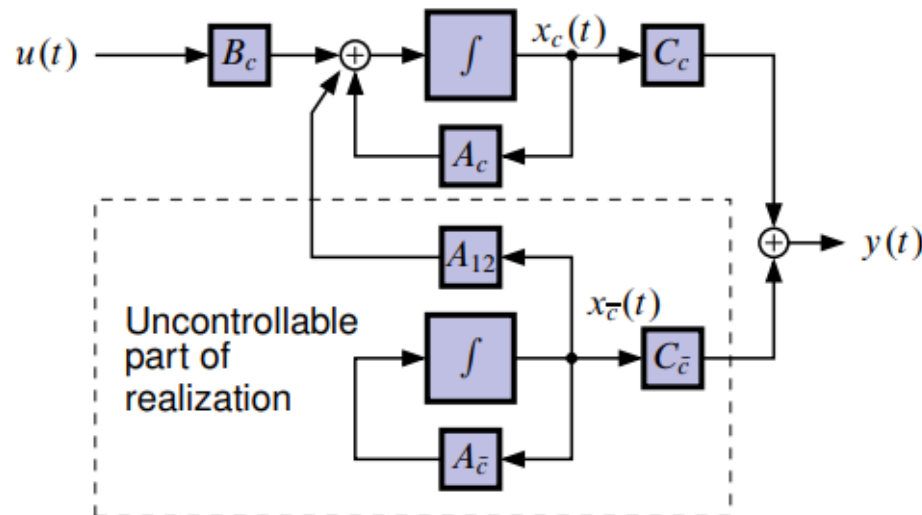
$G(s)$ has n poles unless some cancel with $C \operatorname{adj}(sI - A)B$. Therefore, if $n > k$, $C \operatorname{adj}(sI - A)B$ and $\det(sI - A)$ are not coprime.

Minimal Realization

PROOF: $\text{III} \iff \text{IV}$ Controllable and observable iff minimal.

$\text{III} \implies \text{IV}$: Uncontrollable or unobservable $\implies$ not minimal.

Perform Kalman decomposition to split system into co , $c\bar{o}$, $\bar{c}o$ and $\bar{c}\bar{o}$ parts.



$$\bar{A} = \begin{bmatrix} A_c & A_{12} \\ 0 & A_{\bar{c}} \end{bmatrix}, \bar{B} = \begin{bmatrix} B_c \\ 0 \end{bmatrix}, \bar{C} = \begin{bmatrix} C_c & C_{\bar{c}} \end{bmatrix}, \bar{D} = [0].$$

Same transfer function using A_c, B_c, C_c as $\bar{A}, \bar{B}, \bar{C}$. Therefore uncontrollable and/or unobservable means not minimal.

Minimal Realization

IV $\implies$ III: Non-minimal means uncontrollable or unobservable.

- Suppose $\{A, B, C, D\}$ is non-minimal.

$$\underbrace{C(sI - A)^{-1}B + D}_{n \text{ states}} = \underbrace{\bar{C}(sI - \bar{A})^{-1}\bar{B} + \bar{D}}_{r < n \text{ states}}$$

- $\frac{CB}{s} + \frac{CAB}{s^2} + \frac{CA^2B}{s^3} + \dots = \frac{\bar{C}\bar{B}}{s} + \frac{\bar{C}\bar{A}\bar{B}}{s^2} + \frac{\bar{C}\bar{A}^2\bar{B}}{s^3} + \dots$

- Consider

$$OC = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \begin{bmatrix} B & AB & A^2B & \dots & A^{n-1}B \end{bmatrix}$$

Minimal Realization

$$\begin{aligned}
 &= \begin{bmatrix} CB & \dots & CA^{n-1}B \\ \vdots & & \vdots \\ CA^{n-1}B & \dots & CA^{2n-2}B \end{bmatrix} = \begin{bmatrix} \bar{C}\bar{B} & \dots & \bar{C}\bar{A}^{n-1}\bar{B} \\ \vdots & & \vdots \\ \bar{C}\bar{A}^{n-1}\bar{B} & \dots & \bar{C}\bar{A}^{2n-2}\bar{B} \end{bmatrix} \\
 &= \begin{array}{c} \updownarrow n \\ \left[\begin{array}{c|c} \begin{array}{c} \bar{C} \\ \bar{C}\bar{A} \\ \vdots \\ \bar{C}\bar{A}^{n-2} \\ \bar{C}\bar{A}^{n-1} \end{array} & 0 \end{array} \right] \begin{array}{c} \updownarrow r \\ \left[\begin{array}{c} \bar{B} \quad \bar{A}\bar{B} \quad \dots \quad \bar{A}^{n-1}\bar{B} \\ \hline 0 \end{array} \right] \updownarrow n-r \end{array} \\ \leftarrow r \quad \leftarrow n-r \quad \leftarrow n \end{array}
 \end{aligned}$$

- Therefore $\det(\mathcal{O}) \det(\mathcal{C}) = 0$, so the system is either unobservable, or uncontrollable, or both.
- The four equivalences have now been proven.